

CFPS Manufacturability Predictor

v3.3 trained prediction backend + v2.5 diagnostic/release layer

Release status	HOLD
v2.5 compatibility score	49.1/100
Protein gate score	50.6/100
DNA gate score	45.8/100
v3.3 overall prioritization	49.0/100
Confidence	higher

WARNING: The combined release score is below 85/100. Treat this construct as requiring optimization or additional validation before release.

1. Prediction summary

Endpoint	Prediction
CFPS soluble fraction	43.1%
Operational aggregation / insoluble fraction	56.9%
General solubility probability	54.1%
RP3 production probability - protein	50.3%
RP3 production probability - protein + DNA	45.8%
NESG expression score	3.52/5
NESG solubility score	1.98/5
OOD index	0.43

2. v2.5 compatibility/release logic

The compatibility layer preserves the earlier v2.5 release rule: combined score = 70% protein score + 30% DNA score. GO requires combined ≥ 85 , protein ≥ 72 , and DNA ≥ 78 . When DNA is absent, the result is PROVISIONAL and cannot receive final GO.

Gate	Value	Requirement	Pass
Combined	49.1/100	≥ 85	False
Protein	50.6/100	≥ 72	False
DNA	45.8/100	≥ 78	False

3. Warnings, likely consequences, and corrective actions

Severity	Issue	Evidence / consequence	Recommended action
MEDIUM	Moderate CFPS solubility	Model predicts 43.1% soluble fraction. Soluble yield may be limited even when total synthesis is adequate.	Screen 20-30 C, chaperone condition, and template concentration before redesigning sequence.
MEDIUM	Intermediate recombinant production probability	RP3 production probability is 45.8%. Outcome is uncertain and sensitive to construct/context.	Use a pilot expression screen before committing to scale-up.
MEDIUM	Long DNA homopolymer	Longest homopolymer 7 nt. Long homopolymers can complicate synthesis or sequence stability.	Break the homopolymer with synonymous codon substitutions.

4. Experimental follow-up suggestions

- Quantify total and soluble yield across a small condition matrix.
- Run replicate micro-scale expression under two temperatures.
- Sequence-verify the synthesized template and compare expression after synonymous redesign.

5. Construct diagnostics

Feature	Value
Length	377
Approx. molecular weight	41.9 kDa
GRAVY	-0.113
Hydrophobic fraction	39.8%
Longest hydrophobic run	5
Charge proxy	-0.002
Cysteines	8
Sequence entropy	4.13
DNA GC	51.6%
DNA GC3	59.9%
Longest DNA homopolymer	7

6. Input record and model notes

Protein sequence length: 377 aa

Coding DNA supplied: yes

Construct ID: Demo-01

Expression context: E. coli CFPS / PURE-like

Planned temperature: 30 C

Chaperone condition: Not specified

Model limitations

- Absolute CFPS yield (mg/mL) is not trained in this release.
- CHO secretion/yield is not yet model-backed in this release.
- Aggregation is operationally defined as 1 - eSOL/PURE soluble fraction.
- The v2.5 compatibility scores are a transparent release layer built from v3.3 endpoints; they are not a separately retrained v2.5 ML model.
- The overall score is a prioritization score, not a directly measured endpoint.

Interpretation note: model outputs are prioritization and risk-estimation aids. They do not replace experimental validation. Absolute CFPS yield and CHO secretion/titer are not claimed unless a trained quantitative head is available.

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